$$= \lim_{x\to 0}\left\{\frac{\log(1+x^3)}{x^3}\times\frac{x^3}{\sin^3 x}\right\}$$

$$= \lim_{x\to 0}\frac{\log(1+x^3)}{x^3}\times\frac{1}{\lim\limits_{x\to 0}\left(\frac{\sin x}{x}\right)^3}$$

$$= 1\times\frac{1}{1} = 1$$

Exercise 5.2

1. Evaluate the following

   (i) $\lim\limits_{x\to 2}\frac{x^3+2}{x+1}$

   (ii) $\lim\limits_{x\to\infty}\frac{2x+5}{x^2+3x+9}$

   (iii) $\lim\limits_{x\to\infty}\frac{\sum n}{n^2}$

   (iv) $\lim\limits_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1-x}}{x}$

   (v) $\lim\limits_{x\to a}\frac{x^{\frac{5}{8}}-a^{\frac{5}{8}}}{x^{\frac{2}{3}}-a^{\frac{2}{3}}}$

   (vi) $\lim\limits_{x\to 0}\frac{\sin^2 3x}{x^2}$

2. If $\lim\limits_{x\to -a}\frac{x^9+a^9}{x+a} = \lim\limits_{x\to 3}(x+6)$, then find the values of a.

3. If $\lim\limits_{x\to 2}\frac{x^n-2^n}{x-2} = 448$, then find the least positive integer n.

4. If $f(x)=\frac{x^7-128}{x^5-32}$, then find $\lim\limits_{x\to 2} f(x)$

5. Let $f(x)=\frac{ax+b}{x+1}$, If $\lim\limits_{x\to 0} f(x)=2$ and $\lim\limits_{x\to\infty} f(x)=1$, then show that $f(-2)=0$.

Derivative

Before going into the topic, let us first discuss about the continuity of a function. A function $f(x)$ is continues at $x = a$ if its graph has no break at $x = a$

If there is a break at the point $x = a$ then we say that the function is not continuous at the point $x = a$.

If a function is continuous at all the point in an interval, then it is said to be a continuous in that interval.

5.2.8 Continuous function

A function $f(x)$ is continuous at $x = a$ if

(i) $f(a)$ exists

(ii) $\lim\limits_{x\to a} f(x)$ exists

(iii) $\lim\limits_{x\to a} f(x) = f(a)$

Observation

In the above statement, if atleast any one condition is not satisfied at a point $x = a$ by the function $f(x)$, then it is said to be a discontinuous function at $x = a$

5.2.9 Some properties of continuous functions

If $f(x)$ and $g(x)$ are two real valued continuous functions at $x = a$, then

(i) $f(x) \pm g(x)$ is also continuous at $x = a$.

(ii) $f(x)\, g(x)$ is also continuous at $x = a$.

(iii) $kf(x)$ is also continuous at $x = a$, k be a real number.

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